SecureDT AI Model Installation Guide

Random Forest Risk Model for Task Security Risk Prediction

REQUIRED MODELS

These models are required for the core functionality of SecureDT.

1. Risk Prediction Model (REQUIRED)

Purpose: Predicts the security level of an incoming scheduled task from labelled synthetic workloads.
Package: scikit-learn RandomForestClassifier
Install command: pip install scikit-learn>=1.3
Training command: python scripts/train_ai.py
Dataset: ai_training_10000.csv (synthetic, seed 42)
Input features: cpu, memory, latency, trust, security_risk, attack_severity, privacy, workload, storage_integrity, attack_exposure (10 features)
Output: Risk level (safe/trained/critical) + confidence score (0-1)
Model path: models/risk_model.pkl
CPU/GPU requirement: CPU-only; scikit-learn RandomForest trains efficiently on CPU. No GPU required.

OPTIONAL ENHANCEMENTS

These enhancements improve capabilities but are not required for core operation.

2. Groq Cloud LLM Enhancement (OPTIONAL)

Purpose: Natural-language explanations, experiment summaries, risk commentary via Groq Cloud API.
Package: groq SDK (pip install groq)
Install command: pip install groq
Configuration: GROQ_API_KEY and GROQ_MODEL in .env file
Model selection: GROQ_MODEL=llama-3.3-70b-versatile (default) or mixtral-8x7b-32768 or gemma-7b-it
AI_PROVIDER=groq or hybrid in .env to enable
Fallback: If key missing, app degrades to local AI mode without crash
CPU/GPU requirement: Server-side inference on Groq Cloud; no local GPU needed.

3. SHAP Feature Explanations (OPTIONAL)

Purpose: Detailed feature importance explanations for risk predictions.
Package: shap (pip install shap)
Feature reasons: displayed in UI when shap installed
Fallback: Deterministic rule-based reasons always available
CPU/GPU requirement: Lightweight Python library; CPU only.

4. Risk Model Metrics (AUTOMATIC)

ROC-AUC, F1-score, feature importance, validation results written to models/risk_metrics.json
Automatic with sklearn training
Read via /api/ai/analyze and /api/export/experiments

INSTALLATION STEPS

Step 1: Install dependencies - pip install -r requirements.txt
Step 2: Train the AI model - python scripts/train_ai.py
Step 3: Verify model file exists - ls models/risk_model.pkl
Step 4: Start the application - python manage.py runserver (or python app.py)
Step 5: Test AI analysis - curl -X POST /api/ai/analyze

VERIFICATION

After installation, verify:
• python manage.py check → no issues
• python scripts/setup_check.py → AI dependencies OK
• /api/ai/analyze endpoint returns risk level + confidence
• models/risk_model.pkl exists and is loadable
• Fallback works: delete model file, re-run analysis → rule-based

IMPORTANT NOTES

• The risk model is trained on synthetic data (seed 42) for demonstration purposes.
• Model trains on CPU only; no GPU acceleration required.
• If models/risk_model.pkl is deleted, the engine auto-falls back to deterministic rules.
• Groq LLM enhancement is completely optional; core risk prediction works without it.
• Model retraining: re-run scripts/train_ai.py with updated dataset if needed.